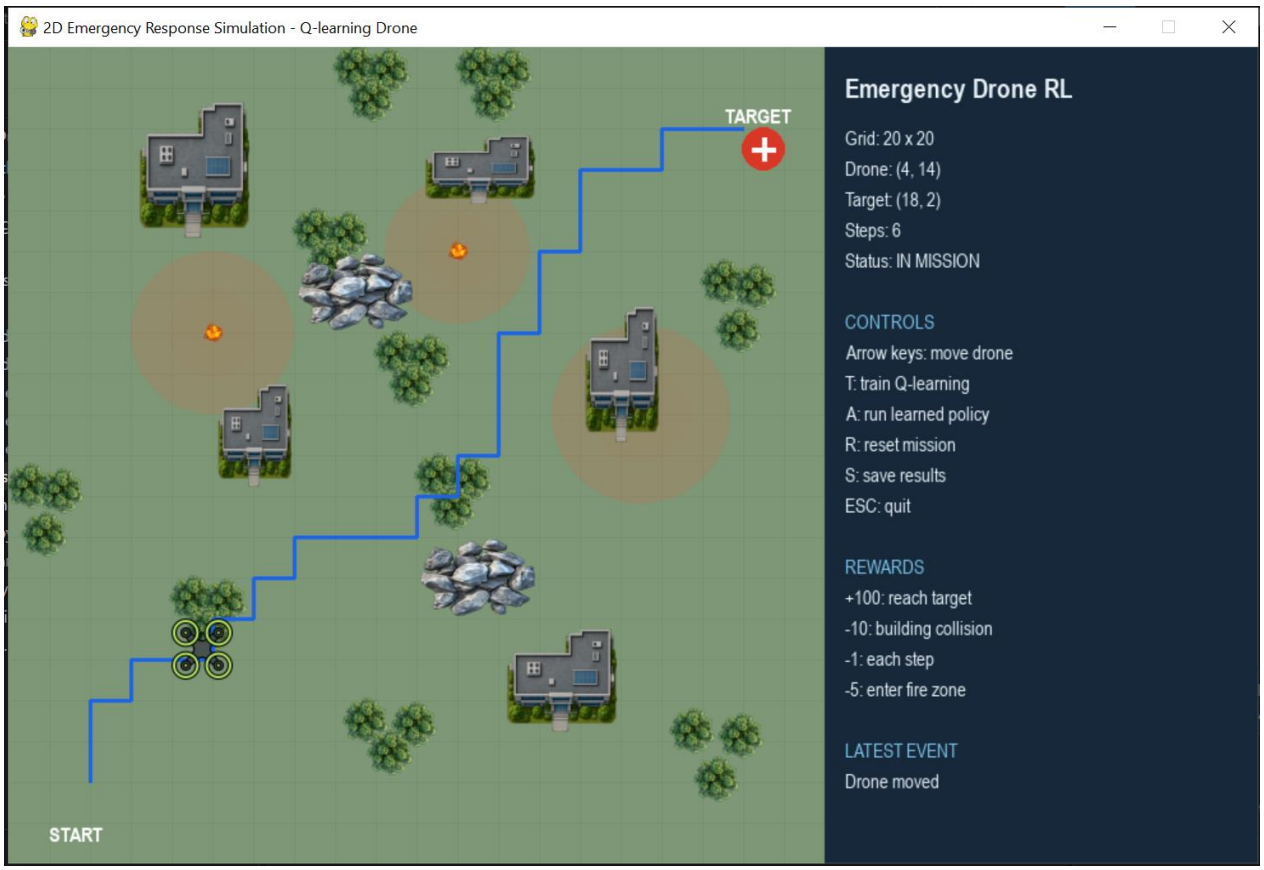
Emergency-Response Drone Simulation Using Reinforcement Learning

Project Summary

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1. Project Aim and Implementation

The aim of this project is to develop a simulated drone agent that learns to navigate an environment and reach an emergency target using Reinforcement Learning (RL).

I developed a 2D simulation using Python and Pygame. The environment contains a grid-based map with buildings, obstacles, roads, fire hazards, a drone agent, and a target location.

The next step in the implementation is to use Q-learning, a model-free Reinforcement Learning algorithm, to help the drone learn navigation through interaction with the environment.

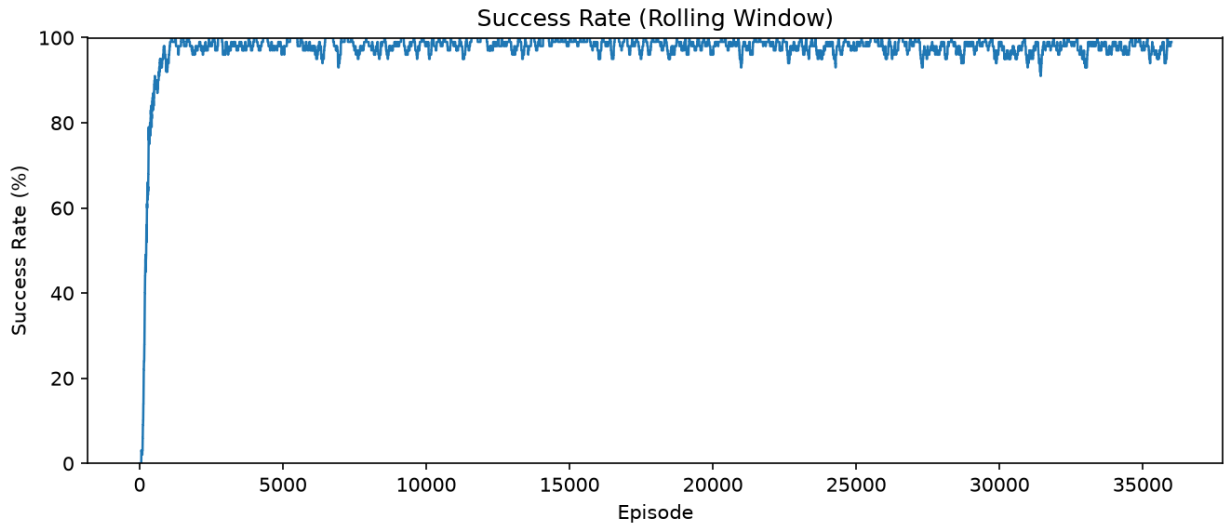
The agent learns through a reward function that encourages reaching the target and penalizes unsafe movements, invalid actions, and hazardous situations. The goal is to

investigate how an agent can learn navigation and decision-making from repeated interactions.

2. Preliminary Results

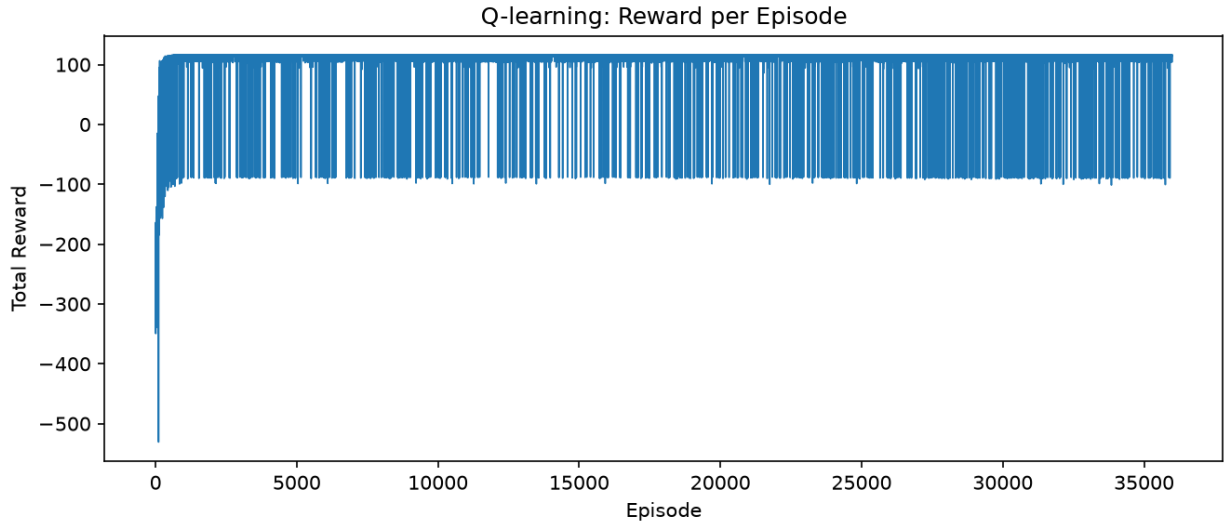
The project includes training logs and visualizations to examine the agent's learning behaviour. The current training log records approximately 97.44% of successful episodes across 36,000 recorded episodes. These results are preliminary and should be verified by reproducing the training run with the corresponding configuration before drawing firm conclusions.

Figure 1. Success Rate During Training



This graph shows how frequently the agent reaches its target over training. A consistently high success rate would indicate that the agent is learning to complete the task reliably.

Figure 2. Episode Reward During Training



The reward graph shows how the total reward changes across episodes. It helps assess whether the agent's behavior improves under the chosen reward function.

3. Limitations and Future Work

The current project focuses on a simplified, single-drone, grid-based environment. It does not yet fully model real-world flight dynamics, battery constraints, uncertain sensing, or changing fire conditions.

My immediate next step is to extend the simulation from a single-target task to a multi-target emergency mission, in which the drone must visit several targets in sequence while avoiding hazards.

I also plan to evaluate mission-completion rate, route length, and hazard collisions. In the longer term, I am interested in multi-drone coordination and intelligent mission planning that considers emergency priorities and safety constraints.

This project has helped me explore state and action design, reward functions, and learning-based autonomous decision-making. I am interested in developing these ideas further and understanding how they could relate to research on intelligent and safety-aware autonomous systems, including possible combinations of learning-based methods with explicit rules.

4. Research Interest

I would welcome the opportunity to discuss this project and explore how my interests in Reinforcement Learning, autonomous systems, and decision-making could connect with the research conducted within the EvoFLIGHT project.